

**Aim: Decision Tree Classifier.**

### 1.Importing the libraries.

```
[14]: from sklearn.datasets import load_iris
      from sklearn.tree import DecisionTreeClassifier, plot_tree
      from sklearn.model_selection import train_test_split
      from sklearn.metrics import accuracy_score
      import matplotlib.pyplot as plt
```

### 2.Loading the iris and training and testing the data.

```
[11]: iris = load_iris()
      X = iris.data           #Features = Training
      y = iris.target         #Labels = Testing
```

### 3.Giving test size and creating the creating a DecisionTreeClassifier model using fit.

```
[31]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=42)
```

```
[32]: model = DecisionTreeClassifier()
      model.fit(X_train,y_train)
```

```
[32]: ▾ DecisionTreeClassifier ⓘ ⓘ
      DecisionTreeClassifier()
```

### 4.Checking the accuracy of the model.

```
[33]: y_pred = model.predict(X_test)
```

```
[34]: accuracy = accuracy_score(y_test , y_pred)
      print("Accuracy of the model:", accuracy)
```

Accuracy of the model: 0.9666666666666667

### 5.Plotting the DecisionTreeClassifier model.

```
[35]: plt.figure(figsize = (12,8))
      plot_tree(model, filled = True, feature_names = iris.feature_names, class_names = iris.target_names)
      plt.show()
```

**OUTPUT:**

